



SSZG622 Software Project Management



BITS Pilani

2025-26 First Semester
Individual Assignment 1

Topic : Schedule Management and Planning
Name : Harmalkar Rahul Rajan Sayali
Student ID : 2024TM93073

I certify that this report submitted towards the Software Project Management course assignment is my own work. I also certify that I have not copied the work of other students or the material from any other sources (books, online documents, etc.).

I understand that if this report scores a similarity score above 25%, zero marks will be awarded for this assignment.

Harmalkar Rahul Rajan Sayali
2024TM93073

Executive Summary



- This report reviews the role of **schedule management and planning** in a mid-sized IT services organization that delivered an enhancement project for a retail e-commerce client.
- The project focused on upgrading the client's order processing platform with improved scalability and seamless payment integration.
- The project team used tools like Microsoft Project and sprint-based tracking to manage timelines. However, execution was affected by delays in client feedback, resource constraints, and overlooked interdependencies. These factors resulted in milestone shifts and additional coordination efforts.
- Despite these difficulties, structured monitoring through dashboards and frequent review sessions enabled the team to regain partial control and finish the project with only minor overruns. The case highlights how estimation precision, stakeholder involvement, and proactive risk planning can significantly strengthen schedule adherence.
- The findings emphasize the value of flexible scheduling approaches that combine structured planning with adaptive methods. Future initiatives can benefit from applying these practices to balance deadlines, resources, and client expectations more effectively.

Real-life Scenario



Company Overview:

- The case study is drawn from a mid-sized IT services provider (referred to as TechNova Solutions) with expertise in enterprise software delivery.
- The company employs around 1,200 professionals and executes projects across sectors including retail, banking, and healthcare.

Project Description:

- The selected project was **an e-commerce system enhancement** aimed at modernizing the client's online ordering solution.
- Major objectives included integrating secure payment gateways and enabling analytical dashboards for decision-making.
 1. Team Size: 35 (mix of developers, testers, analysts, and one project manager)
 2. Planned Duration: 8 months
 3. Delivery Approach: Hybrid Agile, combining upfront scheduling with sprint-based delivery
 4. Planned Milestones: Requirement freeze, sprint reviews, integration testing, UAT, and deployment



How Scheduling Was Applied

The project manager developed a **task hierarchy** through a Work Breakdown Structure (WBS) and mapped it into Microsoft Project for tracking. Dependencies were outlined, and a Gantt chart was used to visualize timelines. Agile ceremonies were included alongside formal planning:

- Three-week sprints with backlog refinement
- Daily stand-ups for task tracking
- Bi-weekly progress reviews with stakeholders Although the schedule was detailed and comprehensive, execution presented unforeseen challenges.

Observed Outcomes

The scheduling framework provided early visibility and clarity, but actual performance deviated from the baseline:

- Requirement Finalization Delays: Client-side approvals took 3 extra weeks.
- Integration Issues: Payment gateway testing was prolonged, postponing downstream activities.
- Resource Bottlenecks: Specialists were allocated across multiple projects, affecting availability.

Ultimately, the project concluded in 9 months, only one month later than planned and within budget, though not without stress on the team.



Key Challenges

1. **Overlooked Dependencies** – Coordination across modules was underestimated.
2. **Insufficient Contingency** – No effective buffer was built for testing and integration phases.
3. **Resource Sharing Conflicts** – Team members with critical expertise were double-booked.
4. **Slow Client Feedback** – Response cycles caused recurring slippages in milestones.

Personal Reflection

- This experience underlined that scheduling success is not only about detailed planning but also about anticipating uncertainties.
- Building flexibility into plans, improving estimation discipline, and involving stakeholders more actively would have prevented much of the delay

Analysis



Weakness 1: Underestimation of Dependencies

The project team underestimated interdependencies between modules (e.g., integration of the payment gateway and analytics dashboard). This caused rework and delays when downstream activities could not begin on time.

- **Impact:** Delayed integration testing and missed milestone deadlines.
- **Why it was a problem:** As per PMBOK, dependencies must be identified and tracked rigorously. Failure to do so creates schedule risks.

Weakness 2: Inadequate Buffering in Schedule

The initial plan had little contingency built into critical phases such as integration and testing. When client-side delays occurred, the lack of time buffers amplified schedule slippages.

- **Impact:** Extended project delivery by one month.
- **Why it was a problem:** PMBOK recommends schedule reserves to manage uncertainties. Ignoring this reduced flexibility.

Weakness 3: Resource Allocation Conflicts

Key technical resources were assigned across multiple projects simultaneously, leading to availability gaps and priority clashes.

- **Impact:** Some tasks remained idle while waiting for expert inputs.
- **Why it was a problem:** Resource leveling and allocation analysis, if applied, could have balanced workload and reduced delays.

Weakness 4: Ineffective Stakeholder Coordination

Client feedback cycles were slower than expected, but no escalation or structured communication plan was in place to mitigate delays.

- **Impact:** Requirements finalization slipped by 3 weeks.
- **Why it was a problem:** Strong stakeholder engagement is a core principle of schedule management. Lack of timely coordination hurt milestone adherence

Recommendations



1. Strengthen Dependency Mapping

- Use dependency matrices and critical path analysis to identify potential bottlenecks early.
- Tools such as MS Project or Jira Advanced Roadmaps can help visualize dependencies.

2. Introduce Schedule Buffers and Risk Contingencies

- Apply the Critical Chain Method (CCM) with project buffers to absorb uncertainties.
- Reserve at least 10–15% contingency for testing and integration phases.

3. Implement Resource Leveling Practices

- Use resource calendars to avoid double-booking critical experts.
- Apply load balancing across teams to ensure smooth availability.

4. Establish a Stakeholder Communication Plan

- Weekly status updates, escalation protocols, and sign-off checkpoints.
- Early alignment ensures faster decision-making and reduced rework. Implementation

Example: For future projects, a “Schedule Risk Review” can be held bi-weekly to track dependencies, buffer usage, and resource load. This proactive approach ensures risks are addressed before they impact milestones

References



- Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide) (7th ed.)*. PMI.
 - Kerzner, H. (2017). *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*. John Wiley & Sons
-

Thank You